

Environment Setup & Documentation

a) Introduction

This section documents the setup of a controlled cybersecurity testing environment using Kali Linux installed on a virtual machine to support security assessment activities for the BSF-Nutrifeed ecosystem. The environment is configured to facilitate penetration testing, vulnerability assessment, and network security analysis aligned with the responsibilities of a Cybersecurity Analyst within the organization.

The virtualized setup ensures that all security experiments and simulated attacks are conducted safely without disrupting live BSF-Nutrifeed operational systems or exposing sensitive data. Key tools configured in this environment support network reconnaissance, traffic monitoring, vulnerability detection, and controlled exploitation testing, enabling comprehensive evaluation of potential security weaknesses within the BSF-Nutrifeed digital infrastructure.

b) System Requirements

- i) A laptop
- ii) VirtualBox software
- iii) Kali Linux ISO file

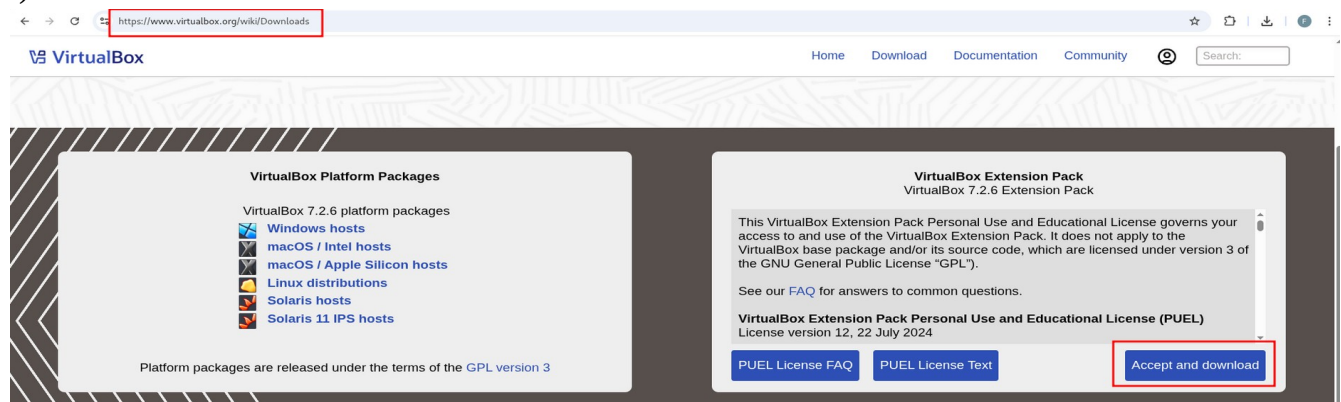
c) Software Downloads

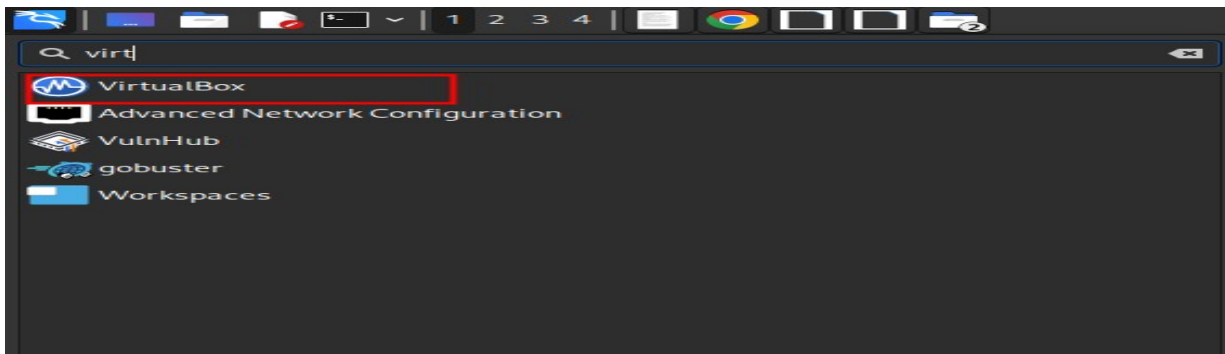
- i) VirtualBox
- ii) Kali Linux ISO file

d) Installation Process

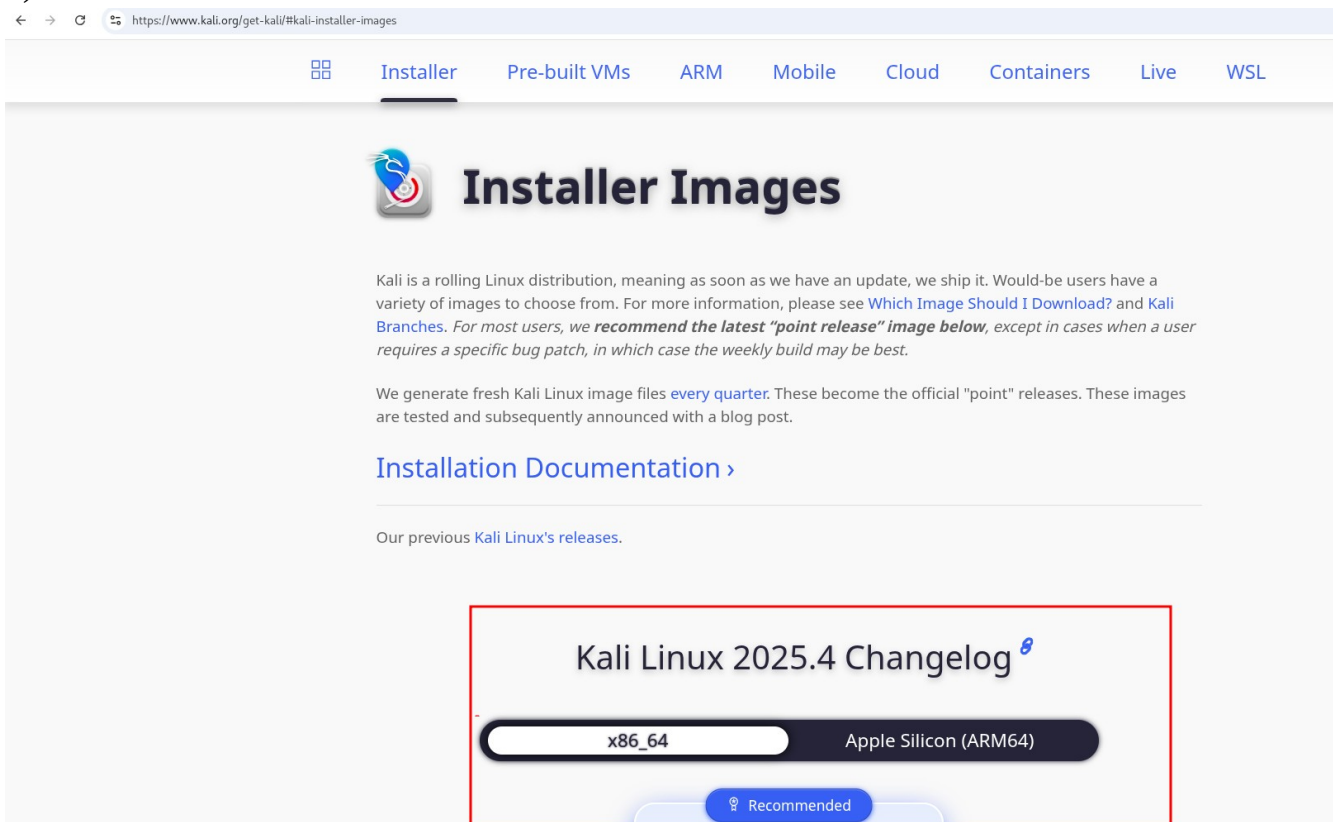
Download the following:

i) VirtualBox





ii) Kali Linux ISO file



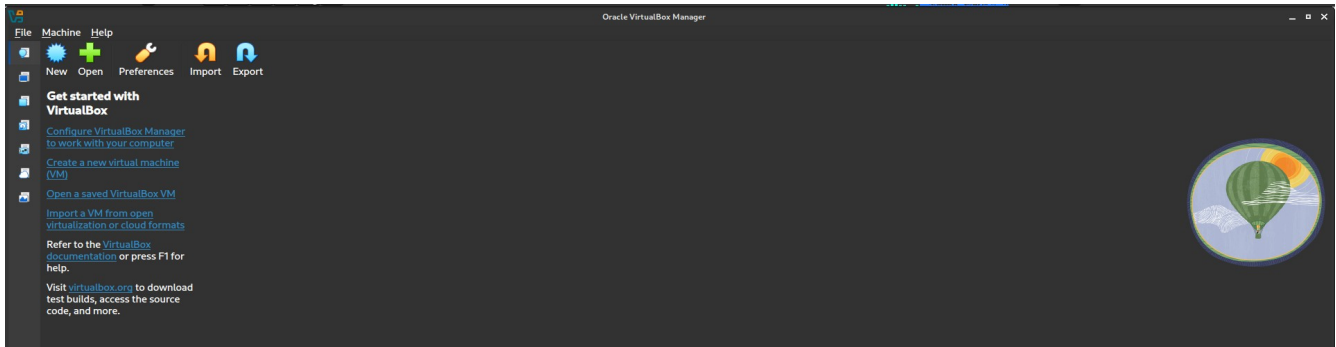
e) Installing Kali linux in VirtualBox

Requirements:

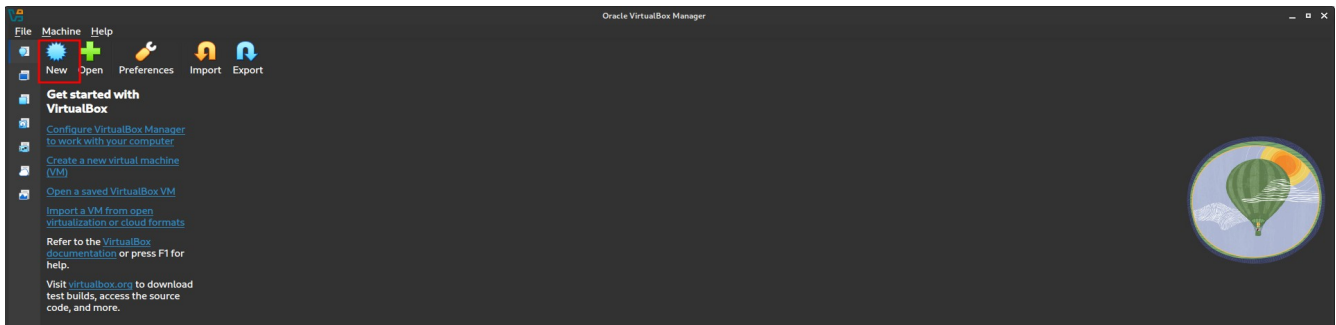
- i) A computer with at least 4GB RAM
- ii) Virtual machine
- iii) Kali Linux ISO file
- iv) Minimum 20GB disk space
- v) Internet connection

Virtual Machine Creation

i) Open the VM

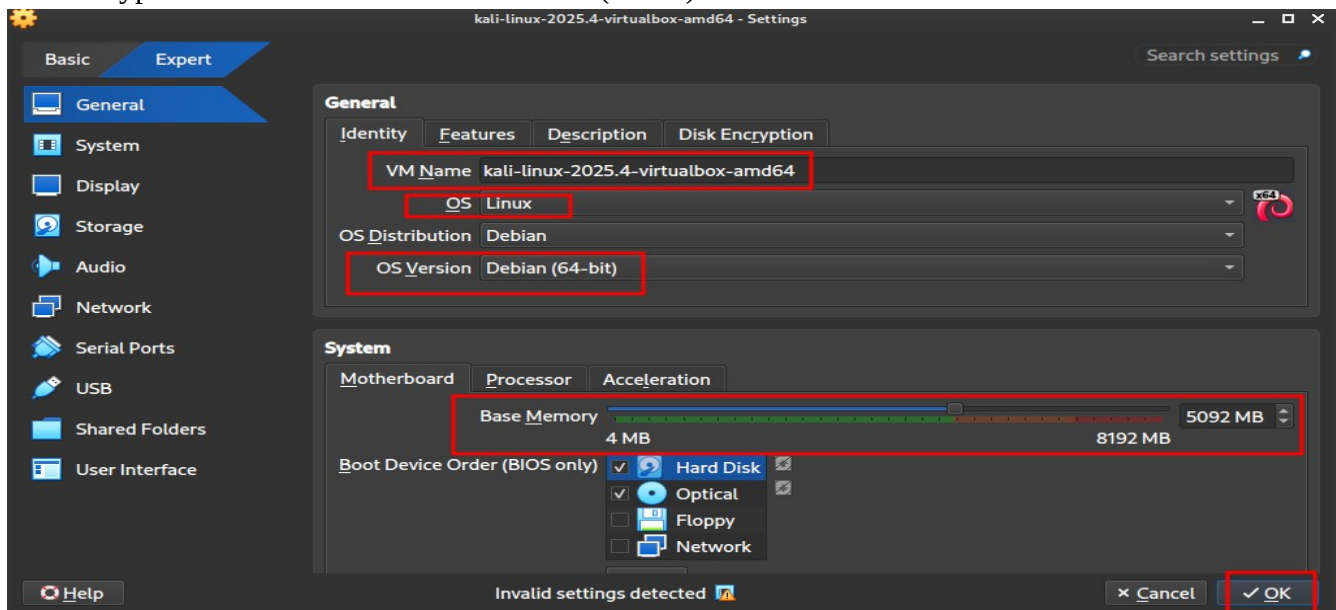


ii) Click new

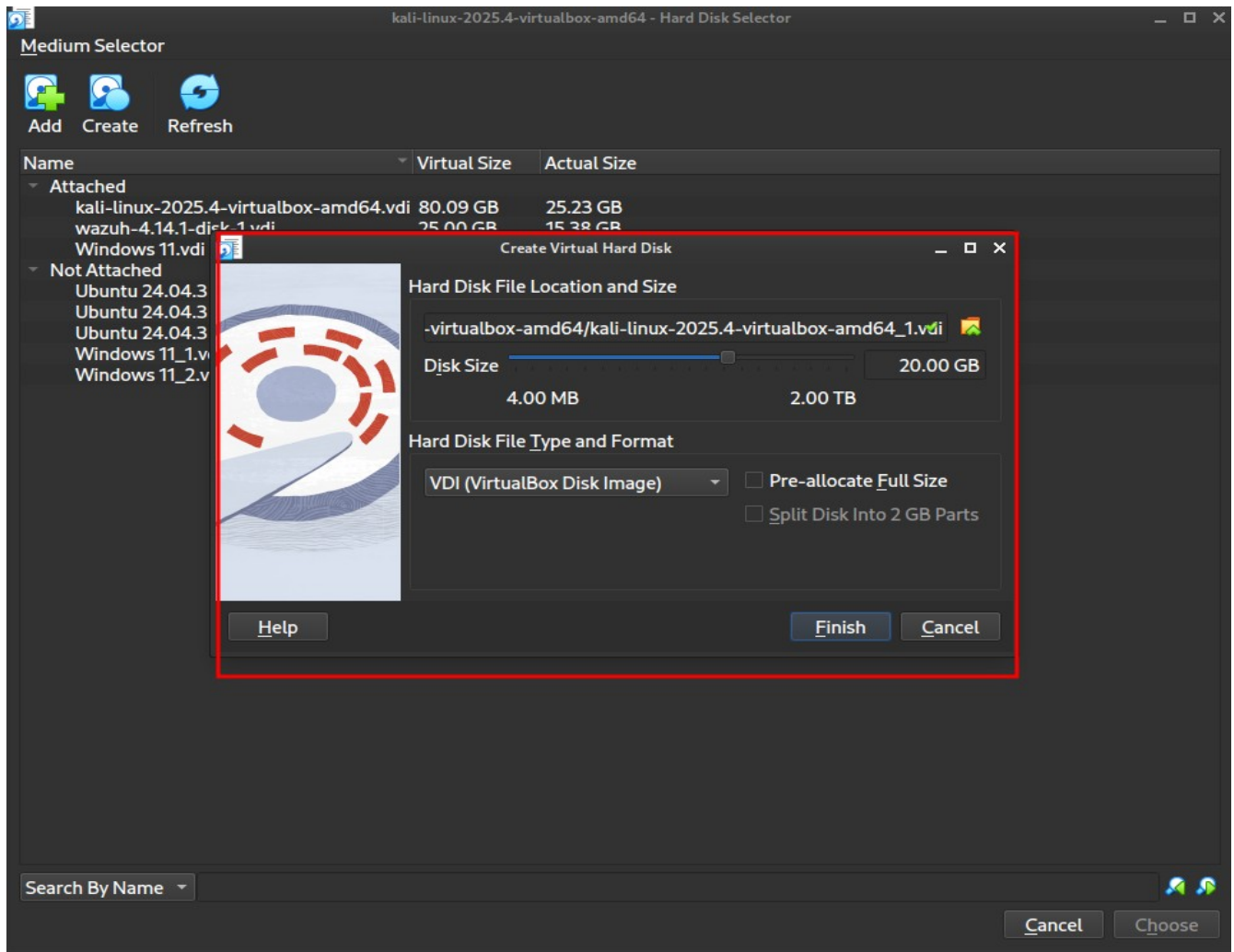


iii) Name the machine and Select:

Type - Linux ==> Version - Debian (64-bit)



iv) Create Virtual Hard Disk

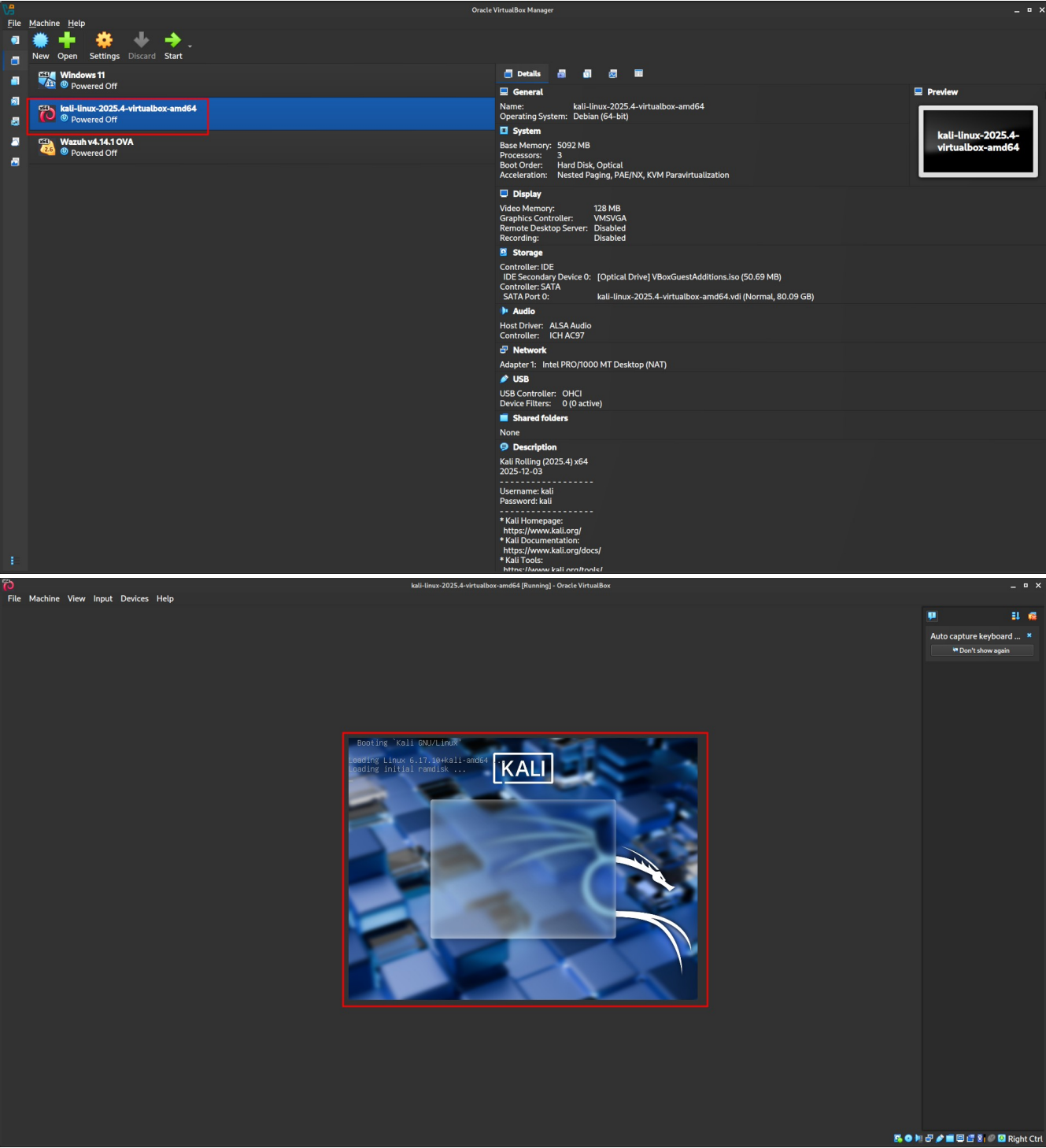


Linux OS installation Process

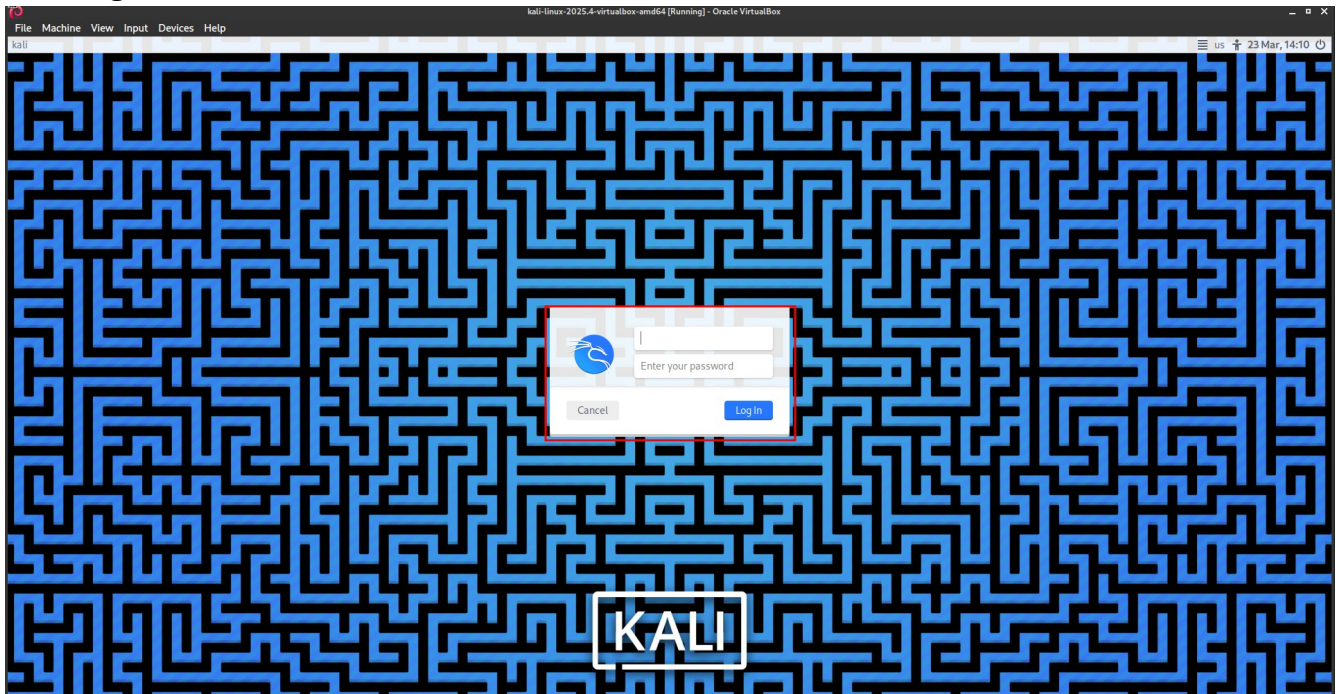
NOTE: OS already installed on my VM

- i) Start VM
- ii) Select Kali Linux ISO as startup disk
- iii) Choose "Graphical Install"
- iv) Select language and region
- v) Create username and password
- vi) Allow installation to complete
- vii) Reboot after installation

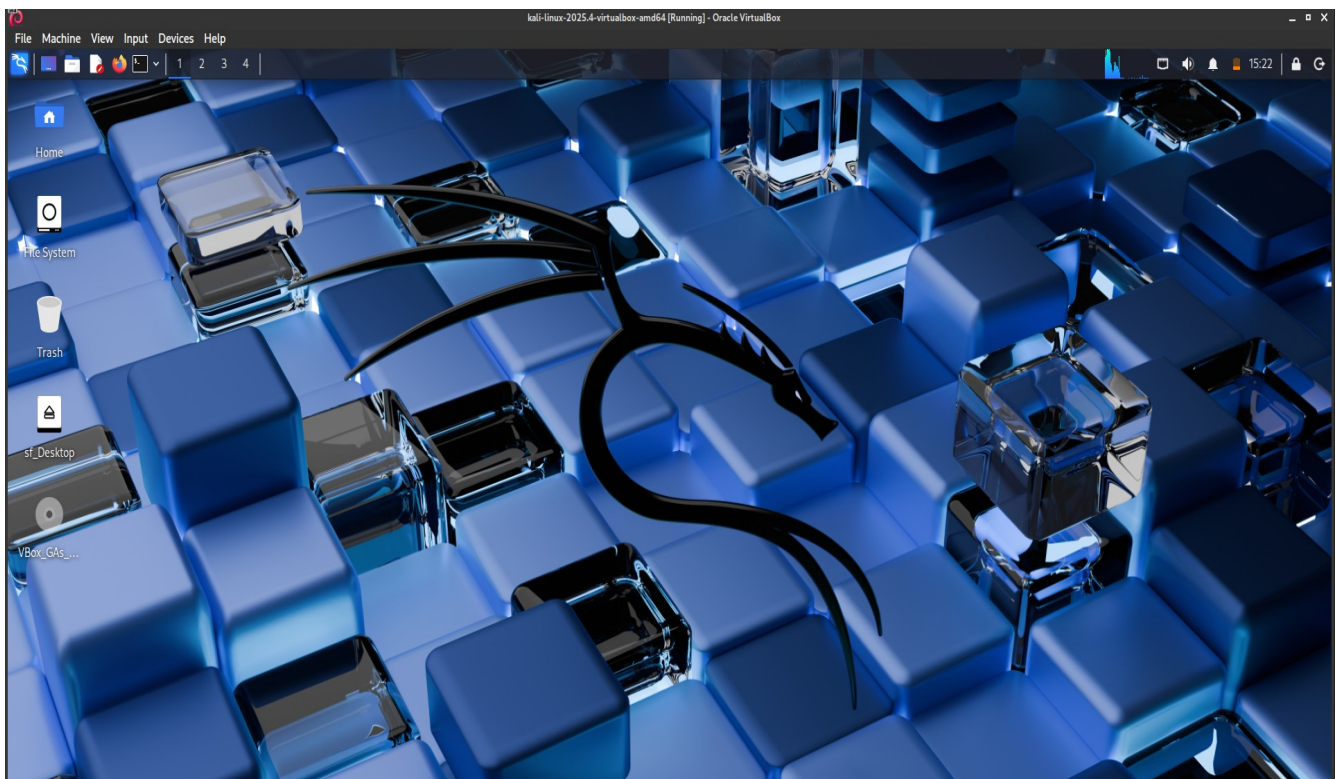
Kali Installer Screen



Kali Login Screen



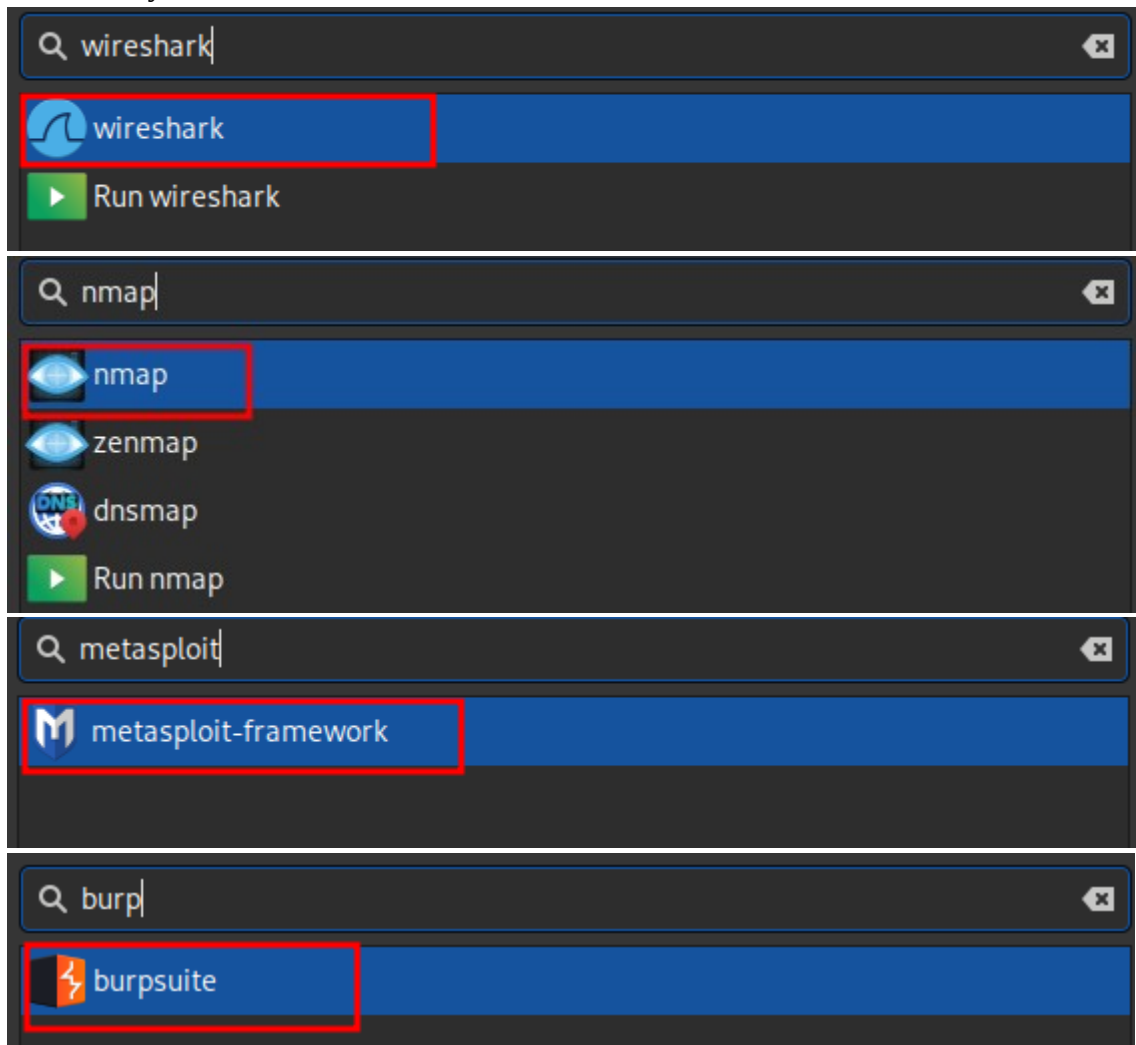
Kali Desktop Interface



f) Post-Installation Verification

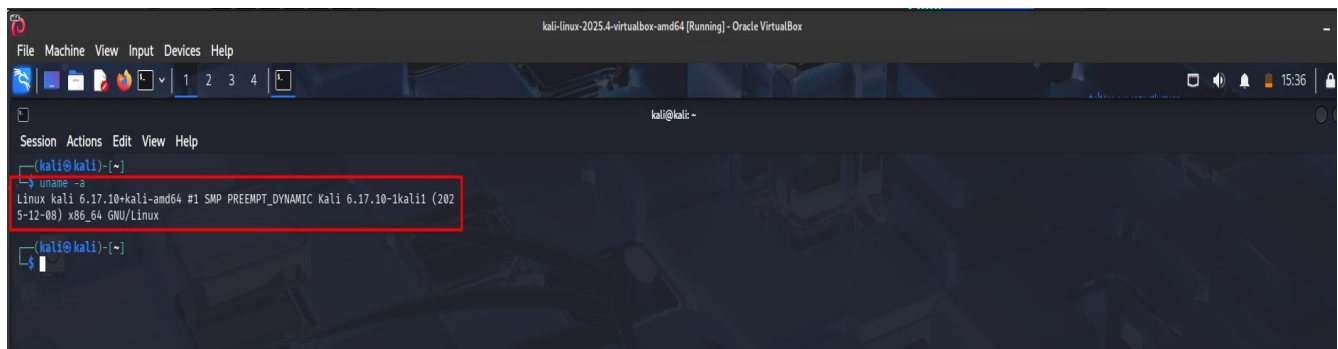
To show Kali works correctly;

i) We open kali “Menu” to confirm that tools (Nmap, Wireshark & Metasploit & Burp Suite) are successfully installed and also run them on terminal for checks;



ii) Let's run:

`uname -a`



ii) We run tool check

nmap --version

wireshark --version

msfconsole --version

Burpsuite --version

Nmap

```
(kali@kali)-[~]  
$ nmap --version  
Nmap version 7.95 ( https://nmap.org )  
Platform: x86_64-pc-linux-gnu  
Compiled with: liblua-5.4.7 openssl-3.5.4 libssh2-1.11.1 libz-1.3.1 libpcap-1.10.5 nmap-libdnet-1.12 ipv6  
Compiled without:  
Available nsock engines: epoll poll select
```

Wireshark

```
(kali@kali)-[~]  
$ wireshark --version  
Wireshark 4.6.0.  
  
Copyright 1998-2025 Gerald Combs <gerald@wireshark.org> and contributors.  
Licensed under the terms of the GNU General Public License (version 2 or later).  
This is free software; see the file named COPYING in the distribution. There is  
NO WARRANTY; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.  
  
Compile-time info:  
  Bit width: 64-bit  
  Compiler: GCC 15.2.0  
  Glib: 2.84.4  
With:  
+brotli  
+Gcrypt 1.11.2  
+GnuTLS 3.8.10 and PKCS#11  
+Kerberos (MIT)  
+libnl 3  
+libpcap  
+libsmi 0.4.8  
+libxml2 2.14.6  
+Lua 5.4.8  
+LZ4 1.10.0  
+MaxMind  
+Minizip 1.3.1  
Without:  
-automatic updates -zlib-ng  
  
Runtime info:  
  OS: Linux 6.17.10+kali-amd64  
  CPU: Intel(R) Core(TM) i5-8250U CPU @ 1.60GHz (with SSE4.2)  
  Memory: 4894 MB of physical memory  
  Glib: 2.86.2  
  Locale: LC_TYPE=en_US.UTF-8  
  Plugins: supported, 0 loaded  
With:  
+brotli 1.1.0  
+c-ares 1.34.5  
+Gcrypt 1.11.3  
+GnuTLS 3.8.11  
+libpcap 1.10.5 (with TPACKET_V3)  
+libsmi 0.4.8  
+LZ4 1.10.0  
+nghttp2 1.64.0  
+nghttp3 1.12.0  
+PCRE2 10.46 2025-08-27  
+Qt 6.9.2  
+QtMultimedia  
+Snappy 1.2.2  
+xxhash 0.8.3  
+zlib 1.3.1  
+Zstandard 1.5.7
```

Msfconsole

```
(kali@kali)-[~]  
$ msfconsole --version  
Framework Version: 6.4.101-dev  
  
(kali@kali)-[~]  
$
```

Burpsuite

```
(oliver@kali)-[~]  
$ burpsuite --version  
[warning] /usr/bin/burpsuite: No JAVA_CMD set for run_java, fall  
ing back to JAVA_CMD = java  
2026.2.3-45662 Burp Suite Community Edition
```